# D2.3 Water potential

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d2-3.html*

*Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 681–690 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## WATER AND SOLUTES

### D2.3.1 Water as a solvent

p. 681

In **solvation**, water molecules surround a solute. Water is polar, so positive ions attract the partially negative oxygen, and negative ions attract the partially positive hydrogen. Polar molecules such as glucose form hydrogen bonds with water.

### D2.3.2 Water moves towards the more concentrated solution

pp. 682–683

| TERM | SOLUTE CONCENTRATION COMPARED WITH THE OTHER SOLUTION |
|---|---|
| HYPOTONIC | Lower |
| HYPERTONIC | Higher |
| ISOTONIC | The same |

Net movement of water is from the hypotonic solution to the hypertonic solution.

### D2.3.3 Water entering and leaving cells

p. 683

Water enters a cell in a hypotonic solution, and leaves a cell in a hypertonic solution.

**Watch out:** in an isotonic solution, water still crosses in both directions, at equal rates. This is a dynamic equilibrium with no net movement.

### D2.3.4 Plant tissue in hypotonic and hypertonic solutions

pp. 683–687

Tissue in a hypotonic solution gains mass and length, and tissue in a hypertonic solution loses them. Calculate **percentage change**, because starting sizes differ. On a graph of percentage change against concentration, the point at zero change gives the **isotonic concentration**. Standard deviation and standard error show how far you can trust the means.

## EFFECTS ON CELLS

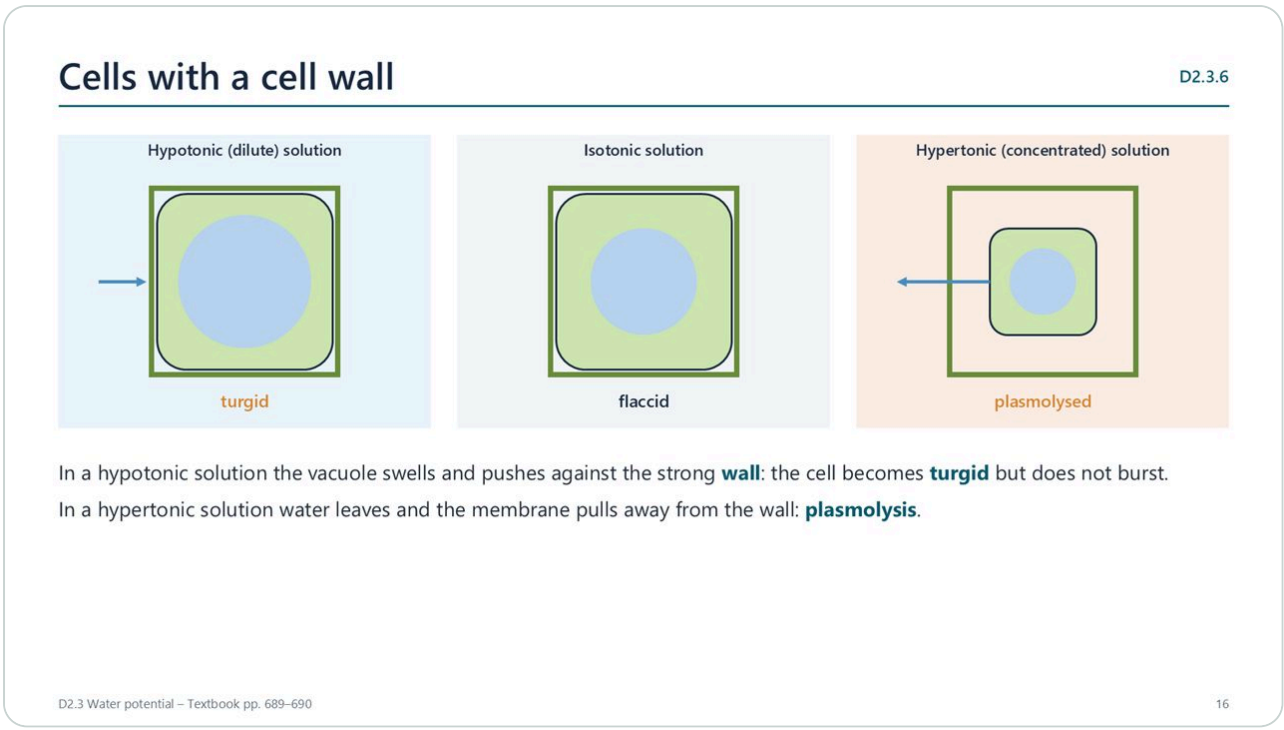
### D2.3.5 Cells without a wall

p. 688

In a hypotonic medium, these cells swell and may burst. In a hypertonic medium, they shrink, with **crenation** in red blood cells. Freshwater unicellular organisms use **contractile vacuoles** to expel water that enters. Multicellular animals keep tissue fluid isotonic with their cells.

## D2.3.6 Cells with a wall

p. 689




In a hypotonic medium, water enters and pressure builds against the wall, giving **turgor pressure**. The wall stops the cell from bursting. In a hypertonic medium, water leaves and the membrane pulls away from the wall, which is **plasmolysis**.

## D2.3.7 Isotonic solutions in medicine

p. 690

Intravenous fluids, such as 0.9% sodium chloride, and solutions for bathing organs before transplant are isotonic. They prevent net water movement, so cells do not swell, burst or shrink.

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